

Kaleb Hood, PhD candidate, Mechanical Engineering

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RG Researchgate-Kaleb Hood

in LinkedIn-Kaleb Hood



Education

- 2023 – expected 2027 > **Ph.D. Mechanical Engineering, Material Science Engineering** Portland State University.
Thesis title: *Development of graphene coating methods for tailored steel surface properties*
Certificate in College Teaching Portland State University
- 2020 – 2022 > **M.Sc. Material Science Engineering** Portland State University.
Thesis title: *Liquid Exfoliation of Multilayer Graphene for Anti-Corrosive Coatings on Additive Manufactured Metals.*
Graduate Certificate in Semiconductor Manufacturing
- 2013 – 2016 > **B.Sc. Biology, Chemistry minor** Warner Pacific University.

Employment History

- 2024 – current > **Adjunct Faculty** Portland State University, Department of Mechanical and Materials Engineering
- 2023 – current > **Graduate Teaching Assistant** Portland State University, Department of Mechanical and Materials Engineering
- 2021 – current > **Graduate Research Assistant** Portland State University, Department of Mechanical and Materials Engineering
- 2017 – 2021 > **Laboratory Coordinator** Warner Pacific University
- 2016 – 2021 > **Adjunct Instructor** Warner Pacific University, Department of Natural Sciences
- 2016 – 2019 > **Teacher** Educational Ministries
- 2014 – 2016 > **Laboratory Assistant** Warner Pacific University, Department of Natural Sciences

Teaching Experience

- 2024 – current > **Adjunct Faculty** Portland State University.
ME 428-528 *Scanning Electron Microscopy for Materials Analysis*
ME 122 *Introduction to Design*
ME 121 *Introduction to Systems and Controls*
- 2023 – current > **Teaching Assistant** Portland State University.
ME 429-529 *Transmission Electron Microscopy for Materials Analysis*
ME 428-528 *Scanning Electron Microscopy for Materials Analysis*
- 2017 – 2021 > **Adjunct Instructor.** Warner Pacific University
CH 211, 212 *General Chemistry*
BIO 210, 202 *General Biology*
BIO 211, 212 *Anatomy and Physiology*
- 2016 – 2019 > **Teacher.** Educational Ministries
9th-10th grade *Physical Science*
7th-10th grade *English Grammar and Writing*
5th-6th *Chess Club*
1th-4th *Lego Robotics*

Research Publications

Journal Articles

- 1 K. Hood, S. Olson, A. Pham, and J. Jiao, "Kelvin probe force microscopy and raman spectroscopy of graphene grown on 1095 steel," *Microscopy and Microanalysis (Accepted)*, vol. 15, no. 17, p. 1367, Mar. 27, 2026, ISSN: 2079-4991. [DOI: 10.3390/nano15171367](https://doi.org/10.3390/nano15171367).
- 2 S. Olson, K. Hood, and J. Jiao, "Afm and raman imaging characterization of defects in commercially available graphene for use in spintronic devices," *Microscopy and Microanalysis (Accepted)*, vol. 15, no. 17, p. 1367, Mar. 27, 2026, ISSN: 2079-4991. [DOI: 10.3390/nano15171367](https://doi.org/10.3390/nano15171367).
- 3 K. Hood, S. Olson, A. Velasquez, and J. Jiao, "Lateral-force AFM study of graphene thickness effects for reduced steel friction," *Microscopy and Microanalysis*, vol. 31, ozafo48.184, Supplement_1 Jul. 25, 2025, ISSN: 1431-9276, 1435-8115. [DOI: 10.1093/mam/ozaf048.184](https://doi.org/10.1093/mam/ozaf048.184).
- 4 K. Hood et al., "Direct graphene growth on low-alloy and mild steel surfaces controlled by carbon solubility and surface microstructural transformations during chemical vapor deposition," *Carbon*, vol. 232, p. 119 801, Jan. 2025, ISSN: 00086223. [DOI: 10.1016/j.carbon.2024.119801](https://doi.org/10.1016/j.carbon.2024.119801).
- 5 M. S. M. Jiban, K. Hood, Z. Templin, J. Jiao, and F. Zhao, "Raman and SEM enabled development of natural organic fructose thin films for resistive switching memory in neuromorphic computing applications," *Microscopy and Microanalysis*, vol. 31, ozafo48.950, Supplement_1 Jul. 25, 2025, ISSN: 1431-9276, 1435-8115. [DOI: 10.1093/mam/ozaf048.950](https://doi.org/10.1093/mam/ozaf048.950).
- 6 S. Olson, K. Hood, and J. Jiao, "High resolution STM images achieved using an open-source microcontroller-enabled tungsten tip etching system," *Microscopy and Microanalysis*, vol. 31, ozafo48.181, Supplement_1 Jul. 25, 2025, ISSN: 1431-9276, 1435-8115. [DOI: 10.1093/mam/ozaf048.181](https://doi.org/10.1093/mam/ozaf048.181).
- 7 S. Olson, K. Hood, O. Zietz, and J. Jiao, "Enhancement of spin transport properties in angled-channel graphene spin valves via hybrid spin drift-diffusion," *Nanomaterials*, vol. 15, no. 17, p. 1367, Sep. 4, 2025, ISSN: 2079-4991. [DOI: 10.3390/nano15171367](https://doi.org/10.3390/nano15171367).
- 8 A. Velasquez, K. Hood, J. Dar, S. Olson, D. Lin, and J. Jiao, "Texturing copper powder via annealing for increased optical absorbance," *Microscopy and Microanalysis*, vol. 31, ozafo48.190, Supplement_1 Jul. 25, 2025, ISSN: 1431-9276, 1435-8115. [DOI: 10.1093/mam/ozaf048.190](https://doi.org/10.1093/mam/ozaf048.190).
- 9 K. Hood, S. Lara, S. Olson, M. Silva, and J. Jiao, "Microscopy of surface microstructural phenomena in 8620 steel after annealing," *Microscopy and Microanalysis*, vol. 30, ozaeo44.300, Supplement_1 Jul. 24, 2024, ISSN: 1431-9276, 1435-8115. [DOI: 10.1093/mam/ozae044.300](https://doi.org/10.1093/mam/ozae044.300).
- 10 S. Lara, K. Hood, S. Olson, H. Wang, and J. Jiao, "Characterization of ball mill applied graphene coatings on 316l steel spheres," *Microscopy and Microanalysis*, vol. 30, ozaeo44.293, Supplement_1 Jul. 24, 2024, ISSN: 1431-9276, 1435-8115. [DOI: 10.1093/mam/ozae044.293](https://doi.org/10.1093/mam/ozae044.293).
- 11 K. Hood, S. Ahmed, and J. Jiao, "Characterization of graphene coatings on 8620 alloy additive manufactured steel," *Microscopy and Microanalysis*, vol. 29, pp. 1436–1437, Supplement_1 Jul. 22, 2023, ISSN: 1431-9276, 1435-8115. [DOI: 10.1093/micmic/ozad067.739](https://doi.org/10.1093/micmic/ozad067.739).
- 12 K. Hood, M. M. H. Tanim, Z. Templin, A. Dao, F. Zhao, and J. Jiao, "Controlled formation of honey carbon nanotube thin films by tailoring the ratio of admixture concentration and annealing time," *Microscopy and Microanalysis*, vol. 29, pp. 146–147, Supplement_1 Jul. 22, 2023, ISSN: 1431-9276, 1435-8115. [DOI: 10.1093/micmic/ozad067.066](https://doi.org/10.1093/micmic/ozad067.066).
- 13 K. Hood et al., "Self-assembly of exfoliated graphene flakes as anticorrosive coatings for additive manufactured steels," *Results in Surfaces and Interfaces*, vol. 11, p. 100 116, May 2023, ISSN: 26668459. [DOI: 10.1016/j.rsurfi.2023.100116](https://doi.org/10.1016/j.rsurfi.2023.100116).
- 14 S. T. Olson, D. Still, K. Hood, O. Zietz, and J. Jiao, "Impact of material and tunnel barrier quality on spin transport in a CVD graphene non-local spin valve device array," *Carbon Trends*, vol. 13, p. 100 300, Dec. 2023, ISSN: 26670569. [DOI: 10.1016/j.cartre.2023.100300](https://doi.org/10.1016/j.cartre.2023.100300).

- 15 M. M. H. Tanim, Z. Templin, K. Hood, J. Jiao, and F. Zhao, "A natural organic artificial synaptic device made from a honey and carbon nanotube admixture for neuromorphic computing," *Advanced Materials Technologies*, vol. 8, no. 14, p. 2 202 194, Jul. 2023, ISSN: 2365-709X, 2365-709X.  DOI: 10.1002/admt.202202194.
- 16 K. Hood, S. Ahmed, R. Schmidt, W. Qian, and J. Jiao, "Investigation of exfoliated graphene flakes and their self-assembly capabilities to form thin coating films," *Microscopy and Microanalysis*, vol. 28, pp. 920–922, S1 Aug. 1, 2022, ISSN: 1435-8115, 1431-9276.  DOI: 10.1017/S1431927622004044.
- 17 W. Qian, R. Schmidt, J. A. Turner, K. Hood, and J. Jiao, "Structural analysis of liquid-exfoliated graphene as building-blocks for anti-corrosion thin films," *Microscopy and Microanalysis*, vol. 28, pp. 980–982, S1 Aug. 1, 2022, ISSN: 1435-8115, 1431-9276.  DOI: 10.1017/S1431927622004263.

Skills

Languages	➤ English (C2), German (B2), Spanish (A2).
Coding	➤ Python, C++, $\LaTeX$
Tools and Microscopes	➤ Chemical Vapor Deposition (CVD), Physical Vapor Deposition (PVD), Chemo-mechanical Polishing (CMP), Electrochemistry, Scanning Electron Microscopy (SEM), Energy Dispersive X-ray Spectroscopy (EDX), Raman Spectroscopy, Optical Microscopy (OM), Atomic Force Microscopy (AFM), Scanning Tunneling Microscopy (STM), Ultraviolet visible and infrared light spectroscopy (UV-Vis-IR).

Other Experience

Outreach

2026	➤ Northwest Science Exposition (NWSE) Judge, Materials and bioengineering, HS special awards
	➤ PSU PhD student Peer mentor program Mentor
2025 – 2026	➤ Mechanical and Materials Engineering Graduate Student Association (MMEGSA) President
2025 – current	➤ Oregon Museum of Science and Industry (OMSI) Volunteer, featured exhibit ambassador
2023 – 2026	➤ PSU Undergraduate Research Mentoring Program Mentor
2021 – 2022	➤ PSU Research Experience Undergraduate (REU) Mentor

Professional Memberships

2026 – current	➤ Order of the Engineer
	➤ Association of Science Communicators (ASC)
2024 – current	➤ American Physical Society (APS)
	➤ American Chemical Society (ACS)
2022 – current	➤ Microscopy Society of America (MSA)

Awards and Achievements

2026	➤ Portland State University, Maseeh College of Engineering and Computer Science (PSU-MCECS) , Outstanding Graduate Student Leadership Award
	➤ Portland State University (PSU) , Lightning Thesis, Excellence in Science Communication <i>winner</i>
	➤ PSU , Three Minute Thesis(3MT), <i>2nd place winner</i>

Other Experience (continued)

- 2024 > **PSU**, Lab Safety Awareness Week Award
- 2024 > **PSU-MCECS**, Outstanding Graduate Teaching Assistant
- 2023 > **Microscopy Society of America (MSA)**, Student Scholar Award
- 2022 > **PSU-MCECS** Outstanding Research Assistant
- 2016 > **Warner Pacific University (WPU)**, Natural Science and Health Departmental Service Award
- 2015 > **Warner Pacific University**, Natural Science and Health Departmental Service Award
- 2013 – 2016 > **WPU Natural Science and Health Scholarship**
- > **WPU Deans Scholarship**
- > **WPU Humanities Department for Arts and Theatre Scholarship**
- 2009 > **Scouting America** Eagle Award

Other Certificates

Presentations

- 2026 > **PSU** Lightning Thesis
- > **PSU** 3MT
- 2025 > **MSA** Poster Session
- 2023 > **MSA** Poster Session
- > **PSU** Young Scientist Sceminar
- 2022 > **MSA** Platform Speaker and Poster Session

References

Available on Request